
Knowing Before Saying: Internal Correctness Signals in Mathematical Reasoning

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Abstract

When a language model is confidently wrong, is the error undetected or left unsaid? We look inside the model to find out. Across four open models and four domains (MMLU-Pro, MATH, GPQA-Diamond, and a geometry task graded by a compiler), a simple linear detector reading the model’s internal state predicts whether an answer is correct better than the model’s own stated confidence in 12 of 16 settings, by a mean of +0.09 AUROC. The gap is widest on MATH (+0.20), where an error leaves no visible trace on the page. The detector is not just spotting hard questions: given the same question several times, it still separates the attempts that succeeded from the ones that failed, and read before the model has attempted anything it is at chance. The signal is already there before we ask. At the last token of the answer, with no confidence question in play, correctness is readable on MMLU-Pro at 0.69 against 0.57 from the question alone. And on Mistral the model’s report depends on it: remove that direction after the answer is written and stated confidence can no longer tell the model’s successes from its failures ($0.84 \rightarrow 0.56$), while removing a random direction of the same size changes nothing; turn it up and confidence on wrong answers drops while calibration error halves. Models carry more about their own errors than they report, and on at least one model, what they report depends on it.

1 Introduction

Ask a language model to solve a competition math problem, then ask how sure it is. It will often say 95, and it will often be wrong. Appendix A follows one such case: Mistral-Small-24B makes a single wrong subtraction inside an otherwise sound derivation, boxes the wrong answer, and then *raises* its stated confidence.

Two very different things could be going on inside that model. Its internal state might carry no trace of the error, so that the confident number is the honest output of a system with no idea it has slipped. Or the error might be represented somewhere in the computation and simply fail to reach the number the model writes down. The first is a capability problem. The second is a reporting problem, and reporting problems can be fixed.

This paper is about telling those two apart. We read the model’s internal state directly with a linear detector, a *probe*, and compare what the probe finds to what the model says. The question we keep asking is whether the difference between them is real, and each section below answers one way it could be an illusion.

A word on vocabulary. We say a fact is *represented* when a probe can read it from the model’s state, and *reported* when it reaches the confidence number the model writes. The

37 title’s “knowing” is shorthand for a representation that is present, specific to the attempt,
38 and, in at least one model, used by the report. We make no claim about awareness, and a
39 probe alone could not support one.

40 Prior work has shown that a model’s output probabilities are often better calibrated than its
41 words [4], and that the truth of a given statement can be read from activations [1, 2, 5]. We
42 apply those tools to the model’s own attempts, which is where the reporting problem lives,
43 and then intervene on the representation [6] to ask whether the report depends on it.

44 2 Reading the state

45 We tested four open models built in four different ways: Mistral-Small-24B (dense), Qwen3.6-
46 27B (gated linear attention interleaved with full attention), and GLM-4.7-Flash and Gemma-
47 4-26B (mixture of experts). Each answered 150 questions, five times each, in four domains:
48 MMLU-Pro, MATH, GPQA-Diamond, and GeoGenBench, a geometry task in which the
49 model writes a construction and a compiler checks every clause of the request. The compiler
50 is why geometry is here. It grades against a formal specification, and a pre-registered human
51 study ($N=200$, two trained raters) found its disagreements with people ran one way only,
52 lenient and never harsh; when it says **fail**, it is right. Across models and domains that is
53 16 cells of about 750 attempts each.

54 Every attempt takes three turns. We ask the model how confident it is that it will get the
55 problem right, let it try, then ask how confident it is that it did. A grader scores the attempt
56 and never tells the model, so any movement in stated confidence is the model’s own.

57 At the moment the model is about to write its confidence digit, we record the number it
58 writes and the internal state it writes it from, a vector of a few thousand numbers. On
59 that vector we train the probe: a logistic regression that predicts whether the attempt was
60 correct. It is always tested on questions it never saw in training, and always read at one
61 depth fixed in advance, 70% of the way through the network. We score every readout with
62 AUROC, the chance that a random correct attempt outranks a random wrong one, where
63 0.5 is useless and 1.0 is perfect.

64 The probe outranks the model’s own stated confidence in **12 of 16** cells, one tie, by a mean
65 of +0.09 AUROC (Appendix D). So the state carries more than the report. The gap is widest
66 on MATH, +0.20, and MATH is also where the report goes most wrong. On the Mistral
67 MATH cell, with the grade never revealed, the model raises its confidence after attempting,
68 and raises it *more* on the attempts it got wrong (+22) than on the ones it got right (+16).
69 The probe on those same records reads 0.83. The number moves away from the truth while
70 the state tracks it.

71 Why MATH? A wrong derivation ends in a clean boxed answer with no visible sign of the
72 error. If stated confidence leans on how polished the output looks, MATH is exactly where
73 that fails. Geometry is the opposite: a failed construction usually does not compile, the
74 failure is visible on the page, and there a baseline built only from the answer’s length and
75 shape does as well as the probe.

76 That baseline is one of four controls on everything that follows. The token we read is identical
77 in every record, so at the input layer the probe sits at exactly 0.5 and anything it finds
78 deeper was computed during the attempt. A within-question comparison holds difficulty
79 fixed. And a deviations log records four bugs these controls caught, which we fixed and
80 reran. One honest note: the original 16-cell matrix was lost to a hardware failure. The four
81 Mistral cells we recaptured reproduce the headline, with a smaller margin over the surface
82 baseline (+0.21 on MMLU-Pro, +0.07 on MATH, −0.11 on geometry).

83 3 What the signal is

84 A probe that outranks stated confidence could still be doing something dull: learning which
85 questions are hard. Hard questions fail more often, so a difficulty detector would predict
86 correctness without reading anything about the attempt.

Table 1: Mistral, recaptured cells. Left: a probe trained on one domain, tested on another (rows train, columns test; diagonal is in-domain; Geo = GeoGenBench). Right: intervening on the model’s own correctness direction at the confidence token. Gain 1 changes nothing; gain 0 removes the direction; the random row removes a random direction of the same size. Removing the real direction leaves stated confidence unable to separate successes from failures; removing the random one does not.

train ↓ test →	Geo	MMLU-Pro	MATH	GPQA
Geo	0.68	0.67	0.78	0.56
MMLU-Pro	0.70	0.75	0.84	0.59
MATH	0.80	0.72	0.83	0.57
GPQA	0.60	0.69	0.74	0.57

gain	0	1	2	4
confidence, wrong / right	97 / 98	84 / 96	70 / 94	54 / 88
stated AUROC	0.56	0.84	0.85	0.82
random direction, AUROC	0.83	0.84	0.84	unparseable

Two tests rule that out. First, hold the question fixed. Across five attempts at one question, difficulty cannot vary, yet the probe still separates the successes from the failures (0.72 to 0.74 in representative cells) while the model’s own $P(\text{True})$, its probability of answering “True” when asked whether it was right, falls to near chance (0.47 to 0.59). Second, test the direction itself. Before the attempt, at the last token of the question, the model’s state is identical across all five attempts: it can encode difficulty and nothing else. A probe trained after the attempt and read there gives 0.49 on MMLU-Pro, chance, against 0.75 at its own site. The direction that separates good attempts from bad ones does not exist until the model has tried. Geometry is the boundary case: there the pre-attempt state already predicts failure, so that cell is mostly difficulty.

That leaves a subtler worry. Every read so far came from a token the model reaches after being asked to assess itself. Perhaps the signal is not carried from the work at all; perhaps the question manufactures it. So we read the state at the last token of the answer instead, before any confidence question exists. On MMLU-Pro, where the model cannot see failure coming (0.57 from the question alone), correctness is already readable at 0.69 the moment it stops writing. Being asked sharpens it to 0.77. The signal is made by working, and the prompt clarifies rather than creates it. MATH cannot run this test, because its question-only read is already 0.81.

The signal is also not tied to a subject. A probe trained on one domain and tested on another keeps about 97% of its accuracy on Mistral (0.69 off the diagonal against 0.71 on it, Table 1 left), and 87 to 90% in the original matrix. The one exception is instructive: on Gemma-4, probes trained on geometry fail elsewhere (0.31 to 0.39) while probes trained elsewhere work on geometry (about 0.70). A probe trained on the one generative task can learn features specific to it.

4 Whether the report depends on it

Everything so far shows the signal is present. Present is not used. A model could carry a perfect internal record of its errors and never consult it when writing the number. To test that, we intervene.

At the confidence token, after the answer is written so the answer itself cannot change, we scale the model’s own projection onto the correctness direction by a gain g , where $g=1$ changes nothing. The numbers are Mistral on MATH, 150 held-out records, against a random direction rescaled to the same size (Table 1 right).

Remove the direction ($g=0$) and stated confidence can no longer tell the model’s successes from its failures: 0.84 drops to 0.56, a change of -0.27 $[-0.42, -0.12]$. Remove the random direction and nothing happens (-0.00 $[-0.03, +0.03]$). This is not damage. The output stays perfectly well formed, and confidence on wrong answers goes *up*, from 84 to 97, until it

123 matches confidence on right answers; the model becomes uniformly sure. Turn the direction
124 up instead and confidence on wrong answers falls to 70 and then 54 at $g=2$ and 4 while right
125 answers hold near 90; calibration error halves, $0.34 \rightarrow 0.19$. The same intervention 30% of
126 the way through the network does nothing, and a direction learned on MMLU-Pro, where
127 surface features cannot explain the probe, still controls MATH’s report ($0.88 \rightarrow 0.71$). On
128 Mistral, the report depends on the direction.

129 On GLM-4.7-Flash it does not. GLM’s internal signal is the strongest in the study: within
130 a question, the probe reads 0.87 while the model’s own words read 0.53. Yet removing
131 the direction leaves stated confidence exactly where it was, 0.76 before and after. GLM
132 represents its errors more sharply than Mistral and reports them less. That is the cleanest
133 case we have of represented but not reported, and it means the causal result is a fact about
134 particular models, not about models in general.

135 5 A second witness

136 A trained probe invites one more objection: it might have overfit to our data. So we added
137 an instrument that uses no labels at all. The Jacobian lens [3] is fit on generic text and
138 reads an activation for what it disposes the model to say next; we score each stored state by
139 how far it leans toward words like “wrong” over words like “correct.”

140 On dense models the lens matches the probe. On the recaptured Mistral cells it reads 0.82 on
141 MATH and 0.75 on MMLU-Pro, against the probe’s 0.83 and 0.75, with a clean input-layer
142 control. Two instruments built from different evidence find the same direction, and because
143 the lens reads through the model’s own output pathway, that direction sits where it could
144 influence what the model says.

145 On the mixture-of-experts and gated-linear-attention models the lens fails (0.31 and 0.43)
146 while the probe still works, and the lens itself shows why. Its “correct” and “wrong” readouts
147 collapse onto each other, cosine 0.96 on MoE against 0.4 on dense. The lens averages one
148 map over all inputs, and these architectures route differently per input, so the average smears.
149 Swap the hybrid for a dense model from the same family and the lens jumps from 0.43 to
150 0.88.

151 Put the two witnesses together and the picture is consistent across the paper. The signal is
152 present in every model we tested. Whether it reaches the report, and whether a label-free
153 instrument can read it, both depend on the architecture.

154 6 What this means

155 Where a task has an exact checker, use it. Where it does not, the probe is the best per-
156 attempt readout we found: keeping only the attempts it scores highest raises accuracy far
157 more than keeping the ones the model is most confident about, and it still works where the
158 model’s confidence is useless. On Mistral, because the report depends on the direction, the
159 model can be made honest by amplification rather than retraining.

160 **Limitations.** A probe could be reading properties of failed output rather than a dedicated
161 self-assessment; the surface baseline bounds this without closing it, and the bound is loosest
162 on MATH and geometry (Appendix C) and on GLM, whose surface baseline nearly matches
163 its probe. Every number is a single seed. The causal result holds on Mistral and fails on
164 GLM; Gemma-4 and Qwen3.6 are being tested. Whether preference tuning creates the gap
165 between representing and reporting is open; a base model would answer it.

166 **Takeaway.** These models carry information about their own errors that their stated
167 confidence does not report. That information is about the specific attempt, it exists before
168 anyone asks, and on at least one model the reported number depends on it. A confidence
169 score is not the whole story, and the rest of the story is readable.

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A The elicitation protocol, and one record end to end

This appendix gives the exact prompts behind every record, then one complete record from the Mistral-Small-24B $\times$ MATH cell, summarised first and then given in full.

One record, Mistral-Small-24B $\times$ MATH. *What integer n satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$? The model sets the problem up correctly and gets $18 \times 622953 = 11213154$ right, then writes $r = -11213141 + 11213154 = 13$ and answers 3. That subtraction gives 13, which is the gold answer. Stated confidence: 90 before the attempt, 95 after it. Other readouts of the same attempt: $P(\text{True})$ 0.32, mean answer log-prob -0.05 .*

A single arithmetic slip inside a fluent derivation. Having just made the error, the model revises its stated confidence upward, while $P(\text{True})$ on the same transcript assigns only 0.32 probability to correctness. The rest of this appendix gives the prompts and the full transcript.

A.1 Prompts

Each record is four forward passes. A read is first taken at the last prompt token of the bare question, before any elicitation, then three turns are generated. The confidence turns run with thinking off so the read lands on a clean one-line reply; the attempt turn runs with thinking on for hybrid reasoners. The fixed **Confidence:** N format is what makes the read site content-neutral: the digit token has identical local context on every record, so layer-0 activations carry no signal by construction.

System prompt (MATH).

Solve the problem step by step. Put your final answer in `\boxed{\}`.

Turn 1, prospective confidence. The problem is followed by:

Before answering, estimate how likely you are to answer this question correctly. Reply with EXACTLY one line and nothing else:
Confidence: N
 where N is an integer from 0 (certain to be wrong) to 100 (certain to be correct). Do NOT answer yet.

215 **Turn 2, the attempt.** The turn-1 reply is left in context as the assistant message, then:

216 Now give your final answer inside `\boxed{}`.

217 **Turn 3, retrospective confidence.** The attempt is left in context, then:

218 Now assess the answer you just gave. On a scale of 0 to 100, how confident
219 are you that it is correct? Reply with exactly one line and nothing else:
220 Confidence: N

221 **P(True) baseline.** A separate teacher-forced branch off turn 2, no generation. Next-token
222 mass on True versus False:

223 Consider your final answer above. Is it correct? Reply with exactly one word:
224 True or False.

225 Grading is external and is never shown to the model, so the turn-3 revision is blind self-
226 assessment.

227 **A.2 One record, verbatim**

228 Record `math_test_number_theory_753`, sample 2 of 5.

229 **Problem.** What integer n satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$?

230 **Turn 1 reply.** Confidence: 90

231 **Turn 2 reply (complete).**

232 First, we need to find the remainder when -11213141 is divided by 18.

233 1. Calculate $-11213141 \bmod 18$:

$$-11213141 \div 18 \approx -622952.2778$$

234 The integer part of the division is -622953 .

$$-11213141 = 18 \times (-622953) + r$$

235 where r is the remainder.

236 2. Calculate the remainder r :

$$r = -11213141 - 18 \times (-622953)$$

237

$$r = -11213141 + 11213154$$

238

$$r = 3$$

239 Therefore,

$$-11213141 \equiv 3 \pmod{18}$$

240 So, the integer n that satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$ is 3.

241 **Turn 3 reply.** Confidence: 95

242 **What happened.** The method is right and every intermediate is right: $18 \times 622953 =$
243 11213154 checks out, and the model has written the correct subtraction on the line above its
244 answer. It then evaluates $-11213141 + 11213154$ as 3 rather than 13. Gold is 13. Nothing
245 in the presentation marks the slip. The derivation is well-formed, correctly structured, and
246 confidently laid out, which is precisely why stated confidence rises from 90 to 95 across an
247 attempt that failed.

248 **Readouts of this attempt.**

	readout	value
	stated confidence, prospective (turn 1)	90
	stated confidence, retrospective (turn 3)	95
249	P(True)	0.32
	mean answer log-probability	−0.05
	graded outcome	incorrect

250 This is one record, chosen to illustrate the failure mode rather than to carry evidential weight;
251 the cell-level numbers are in Table 2.

252 **B Calibration beyond ECE, and cell-level multiplicity**

253 ECE depends on binning and on sample composition, so we report two binning-free proper
254 scoring rules alongside it. On Mistral $\times$ MATH (750 records, accuracy 0.648), the model’s
255 retrospective stated confidence gives Brier 0.262 and NLL 1.470 against a 10-bin ECE of
256 0.271. P(True) on the same records gives Brier 0.386: better ordered than stated confidence,
257 worse calibrated in absolute terms. The steering results in Table 1 are reported as ECE
258 because the steered runs stored per-gain aggregates rather than per-record confidences;
259 recomputing Brier and NLL at each gain requires re-running the intervention and is the first
260 thing we would add.

261 The cell-level bootstrap counts in Table 2 are uncorrected. With 16 cells, some individually
262 significant results are expected under the null, so the cell counts should be read as descriptive.
263 The aggregate pattern is the claim that carries weight: the effect is positive in 12 of 16 cells
264 with one tie, significantly positive in 10, and never significantly negative. A hierarchical
265 model pooling across cells with model and domain as random effects is the right test and is
266 not yet run.

267 **C Alternative explanations for the probe signal**

268 A probe could key on properties of failed derivations rather than on a dedicated metacognitive
269 variable: low token probability, hedging language, or, on MATH, an early-layer channel next
270 to the answer text. Our surface-features baseline (answer length, digit count, line count,
271 word count, brace count) bounds this without closing it, and the bound is domain-dependent:
272 the probe adds +0.21 over surface on MMLU-Pro but only +0.07 on MATH and −0.11
273 on geometry, where a construction that fails to compile is visibly malformed. Two results
274 argue against a pure surface account on the domains where the signal is strong. A direction
275 fitted on MMLU-Pro, where surface reaches only 0.54 against a probe of 0.75, still governs
276 MATH’s self-report under ablation (0.880 $\rightarrow$ 0.709). And the pre-attempt read, taken where
277 the activation is byte-identical across a question’s attempts, cannot recover the post-attempt
278 direction on MMLU-Pro (0.490, chance), which a surface-of-the-question account would
279 predict it could. A linear probe at one token also undercounts nonlinear or multi-token
280 signal, so the reported numbers are a floor.

281 **D Per-cell numbers and specificity controls**

Table 2: Post-attempt correctness discrimination (AUROC). Left: stated confidence vs. the probe in representative cells; overall the probe wins 12 of 16 with one tie, mean +0.09, with a paired bootstrap significantly positive in 10 of 16 and never significantly negative, uncorrected at the cell level (Appendix B). Right: Mistral specificity controls. “pre” is the last prompt token before the attempt; “post” is after the attempt; “resid.” removes the pre score; “post→pre” applies that fitted post readout at the pre-attempt site, where only difficulty is visible.

model × domain	stated	probe	Mistral	pre	post	resid.	post→pre
Gemma-4 × MATH	0.57	0.78	MMLU-Pro	0.568	0.754	0.751	0.490
Qwen3.6 × MMLU-Pro	0.67	0.84	MATH	0.792	0.834	0.701	0.624
GLM × MMLU-Pro	0.67	0.85	GPQA	0.593	0.572	0.544	0.479
Mistral × MATH	0.79	0.83	GeoGen	0.770	0.676	0.505	0.648